

Harshavardhan Gudem

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Summary

Full-stack Software Engineer with 4+ years of experience delivering scalable, cloud-native platforms and developer tooling across e-commerce, education, and enterprise domains. Specialized in Java (Spring Boot), React.js, Docker, Kubernetes, AWS, and Terraform, with strong expertise in ephemeral environments, CI/CD automation, observability, and build/test acceleration. Recently integrated LLM-based APIs to create AI-powered chatbots, demonstrating practical application of GenAI to improve developer and user productivity. Proven ability to accelerate delivery cycles, improve system performance, and build secure, production-grade APIs and platforms aligned with best practices in SDLC, SRE, and DevOps.

Work Experience

Pagoda Tree Education Ltd – Remote

Salt Lake City, Utah, USA

Software Engineer

Jun 2025 – Present

- Migrated a monolithic WordPress education site into a React + Spring Boot microservices architecture, eliminating bottlenecks and improving feature delivery cycles by 40%, enabling faster innovation and maintainability.
- Engineered and deployed RESTful APIs with Spring Boot and PostgreSQL, introducing indexing and caching that reduced API response latency by 35% and improved data consistency across modules.
- Built and deployed a student-facing AI chatbot using LLM-based APIs, enabling real-time Q&A, course tracking, and personalized study support. Improved student engagement and retention by providing always-available, assistance.
- Rebuilt key user-facing modules (registration, course enrollment, payment flows) using React, Redux, Tailwind CSS, resulting in a 25% uplift in satisfaction scores measured by session duration and reduced bounce rates.
- Containerized services with Docker and deployed on AWS EKS (Kubernetes), achieving 99.9% uptime and enabling zero-downtime releases through rolling updates and blue/green strategies.
- Designed secure PostgreSQL schemas and automated migration from MySQL, improving database performance by 30%.
- Built CI/CD pipelines with GitHub Actions, automating builds, tests, and deployments to cut deployment time from hours to minutes, boosting developer velocity.
- Implemented OAuth2 and JWT authentication for robust identity and access management, strengthening compliance with industry security standards.
- Collaborated in Agile ceremonies, improving sprint predictability and accelerating feature delivery by 20%.

Monument Solar - Remote

Salt Lake City, Utah, USA

Software Engineer

Jan 2024 – Jun 2025

(Crashpad: consumer web platform connecting RV travelers with private landowners)

- Designed and scaled Crashpad, a full-stack booking platform built with Spring Boot, Oracle DB, and React, onboarding 15,000+ active users within six months and proving scalability of a new consumer-facing product.
- Engineered high-performance REST APIs for bookings and transactions, applying indexing, query tuning, and pagination to cut backend latency by 30% and ensure smooth user experience.
- Built map-based search, interactive listings, and dynamic filters using React, Redux Toolkit, and Leaflet.js, improving navigation and boosting booking conversions by 22%.
- Implemented secure authentication and role-based access control with OAuth2 + JWT, reducing unauthorized access incidents by 90% and ensuring user trust.
- Automated deployments with AWS (EC2, S3), Kubernetes, and Helm, achieving 99.9% uptime and ensuring zero rollback failures across environments.
- Built CI/CD pipelines (Jenkins + GitHub Actions) with automated testing and observability checks, reducing release cycles by 60% and increasing deployment reliability.
- Delivered a property and booking management system with audit logging, lowering admin errors by 40% and improving operational transparency.
- Documented APIs and workflows in Confluence and Postman, improving onboarding speed and collaboration by 25%.

Teaching Assistant (TA)

Salt Lake City, Utah

University of Utah

Aug 2023 – Aug 2024

- Assisted in delivery of courses on Java, REST APIs, SQL, and SDLC practices to 100+ students across 3 semesters, reinforcing industry-ready development skills.
- Facilitated weekly coding labs and debugging workshops, increasing assignment completion rates by 25%.
- Provided over 300 one-on-one support sessions, clarifying API design, database normalization, and debugging strategies.
- Graded and reviewed capstone projects, providing feedback that improved student code quality and testability by 20%.
- Mentored students on Git workflows and Agile, 80% reported higher confidence in production-grade development.

Total Infra & Mining Solutions

Hyderabad, India

Software Engineer

Nov 2021 - Jul 2023

- Developed backend services for a procurement management platform using Java, Spring Boot, and Hibernate, improving throughput by 20% and reducing processing delays.
- Built secure REST APIs for procurement workflows and vendor systems, with service-layer optimizations that cut response times by 30%, enhancing user efficiency.
- Modeled and optimized schemas in Oracle DB with PL/SQL procedures and triggers, ensuring reliable high-volume data transactions.
- Implemented enterprise-grade security with Spring Security (OAuth2, RBAC), preventing unauthorized access and ensuring compliance with IT controls.
- Containerized services with Docker and deployed to AWS EC2/S3, reducing environment drift and deployment failures.
- Automated CI/CD pipelines in Jenkins (integrated with Maven, Docker, Kubernetes), reducing manual intervention and increasing release reliability by 50%.
- Collaborated in Agile sprints, code reviews, and backend performance audits, driving continuous improvement in scalability and reliability.

South Central Railway,

India

Information Technology Intern

Jun 2020 - Jan 2021

- Developed and maintained responsive front-end user interfaces for a mobile railway ticketing system using HTML, CSS, and modern JavaScript concepts, improving system responsiveness and user satisfaction by 35%.
- Provided frontline support and technical troubleshooting for UI navigation and performance optimizations, resolving issues efficiently and enhancing team collaboration.

Technical Skills

Languages: Java, Python, JavaScript/TypeScript, PHP, SQL, Scala (familiar)

Frameworks: Spring Boot, Hibernate, React, Redux, Node.js, Angular

Cloud: AWS (EC2, EKS, RDS, S3, ELB), Docker, Kubernetes, Terraform, Helm, Jenkins, GitHub Actions, GitLab CI/CD

Databases: PostgreSQL, Oracle, MySQL, MongoDB, Redis

Build / SDLC Tools: Maven, Gradle, Ant, Bazel, Git, Jira, Confluence, Postman

AI / LLMs: LLM API integration (chatbots, RAG workflows, summarization, content generation), prompt engineering, applied GenAI for developer/student productivity

Testing: Logging/metrics pipelines, Selenium, Cypress, Playwright, unit/component/integration testing frameworks

Core Strengths: Ephemeral environments, Build/test acceleration, Developer tooling, Secure APIs, Role-based access control, Linux administration, System design

Education

University of Utah, David Eccles School of Business

Salt Lake City, UT

Master of Science in Information Systems

Aug 2023 - Aug 2024

Jawaharlal Nehru Technological University Hyderabad, CMR Technical Campus

Hyderabad, India

Bachelor in Mechanical Engineering

Jun 2017 - Jul 2021

Projects

Crashpad (Capstone → Scaled at Monument Solar)

Team size: 4 | Role: Frontend Lead & Backend Contributor | Awarded Best Project & Best Live Demo

- Originated the concept for Crashpad and translated it into a working product by designing initial wireframes, user flows, and database schemas, laying the foundation for a scalable full-stack RV parking and booking platform.
- Built the platform with React, Tailwind CSS, Spring Boot, and PostgreSQL, enabling property registration, bookings, and secure payments; processed 500+ test transactions during faculty evaluations.
- Delivered map-based search with OAuth/JWT-secured sessions, reducing login/session-related failures by 25%.
- Contributed backend APIs and optimized PostgreSQL queries, achieving 35% faster response times for booking and search workflows.
- After recognition at the university, the platform was adopted by Monument Solar's innovation group, where it was scaled to 15,000+ production users and evolved into a commercial-grade application.